

Refilling the Buffer in a Storm

The Netherlands' countercyclical capital buffer, 0% → 2% (2022–2024)

Zheng Huang · Diego Gutiérrez

Micro- & Macroprudential Management · Country implementation of a CCyB increase

Master of Financial Technology · Frankfurt School of Finance & Management

October 2026

In 2022–23 the Netherlands raised its countercyclical capital buffer (CCyB) to 2% while the rulebook said 0%. It was refilling a buffer, not fighting a boom

SUMMARY

Decision: CCyB 0% → 1% (ann. May 2022) → **2%** (ann. May 2023, binding 31 May 2024), together with lower buffers for systemically important banks (O-SII).

Puzzle: the Basel reference indicator, the credit-to-GDP gap (credit above its long-run trend), stood at **−33pp** and **−47pp** at the two decisions → the buffer guide implied **0%**.

Q1 · Environment

Recovered economy, then war, energy shock and **11.6%** inflation; ECB hikes from Jul-2022.

Credit cold; housing hot (+19% y/y), then falling.

Banks strong (core capital ratio, CET1, 17.7%) and more profitable as rates rose.

Q2 · Criteria & risks

DNB framework (2022): 2% = “**standard risk environment**”, calibrated on past crisis losses (€12bn), not on credit growth.

Insures against shocks no indicator predicts; housing left to other tools.

Keeps the March-2020 promise to restore releasable capital.

Q3 · Implications

Cheap: ≤ about one fifth of big banks’ CET1 headroom.

A swap: structural buffers cut in 2020 and 2024.

No sign of contraction in credit or its price.

Complements ECB tightening; reciprocated abroad; **€6.7bn** releasable.

Verdict: **normalisation, not tightening.** **Trade-off:** relies on discretion and credibility; release never tested.

Source: DNB (2020, 2022, 2023, 2026); DNB Financial Stability Reports 2022–23; BIS credit-to-GDP gaps; ECB; bank disclosures; own calculations.

In March 2020 COVID struck and neighbours released their buffers, but the Dutch buffer was empty

CCyB release announcements, March–April 2020



Netherlands: CCyB at 0%, nothing to release

DNB improvised (17 Mar 2020):

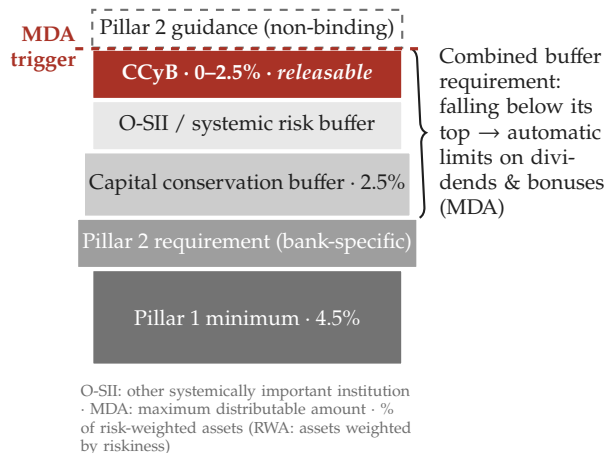
- cut systemic buffers of ING, Rabobank, ABN AMRO: 3% → 2.5 / 2 / 1.5%
- postponed a planned floor on mortgage risk weights
- €8bn freed, up to €200bn of lending

... and made a promise

“Once the situation is back to normal, DNB will compensate the systemic buffers reduction by gradually increasing the countercyclical capital buffer to 2%.”

Source: ESRB, CCyB rates table (release announcements, Mar–Apr 2020); DNB press release “DNB lowers bank buffer requirements to support lending”, 17 Mar 2020.

That mattered because the CCyB is the only layer of bank capital an authority can release in a crisis



Design

- 0–2.5% of risk-weighted assets, on *domestic* exposures
- Increases bind after 12 months; releases are immediate
- EU reciprocity: foreign banks lending in NL hold it too

Why “releasable” matters

- In COVID, banks close to the MDA trigger **cut lending** even when *allowed* to use buffers (stigma, payout limits)
- Only an explicitly **released** buffer is truly usable

Source: CRD (Arts. 130–136); BCBS (2024) d585; Couaillier, Lo Duca, Reghezza & Rodriguez d’Acri (2022), ECB WP 2644. Stack not to scale.

Two years later DNB refilled it in the middle of a storm and against its own rulebook. Was that the wrong moment?

THE QUESTION

The storm (2022–23)

War in Ukraine · record energy prices
HICP inflation **11.6%** (2022)
ECB hikes from **21 Jul 2022**
House prices turning down
Basel gap **−33 / −47pp** → guide 0%



The decision

CCyB **0% → 1% → 2%**
announced May 2022 and May 2023
binding May 2023 and May 2024

Raising capital in a downturn amplifies the cycle. That **procyclical mistake** is what the CCyB was invented to avoid.

Tightening into a slowdown? Fighting a hidden boom? Or something else?

Act 1 · The setting
(Q1 Environment)
timeline · macro
credit & housing · banks



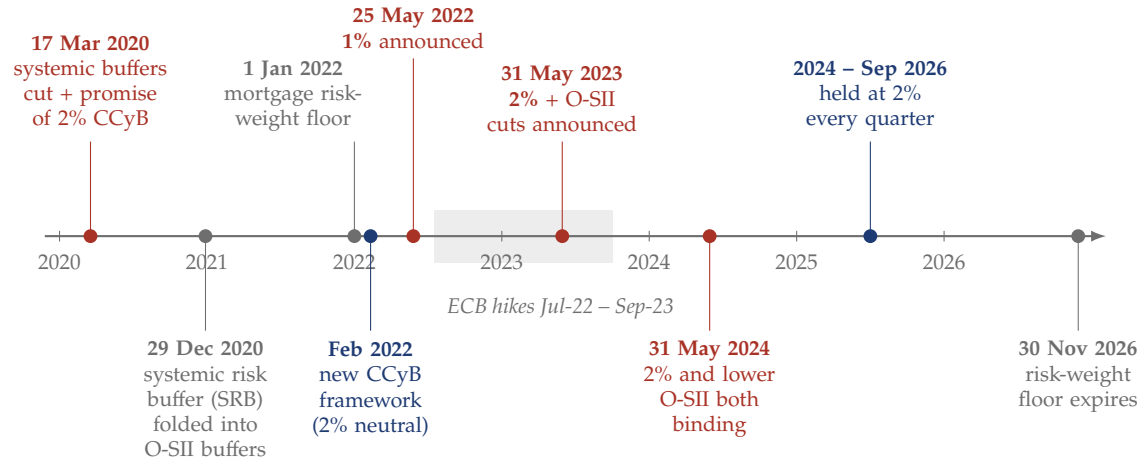
Act 2 · The decision
(Q2 Criteria & risks)
rule vs framework · DNB's reasons
risks & tools · verdict



Act 3 · The consequences
(Q3 Implications)
banks · borrowers · monetary policy
borders · resilience · critiques

Source: CBS (HICP 2022); ECB monetary policy decision, 21 Jul 2022; BIS credit-to-GDP gaps (latest data at each decision: 2021Q4, 2022Q4).

The refill followed the path promised in 2020, one step per year, just as the ECB started to tighten



Source: DNB press releases (17 Mar 2020; 27 May 2022; 31 May 2023; 17 Sep 2026; 24 Apr 2026); ESRB O-SII notifications (27 Nov 2020; 31 May 2023); ECB (21 Jul 2022; 14 Sep 2023).

By then the economy had fully recovered, but war, energy prices and inflation were darkening the outlook

Q1 · ENVIRONMENT

2021Q3

GDP back above
pre-COVID level

11.6%

HICP inflation, 2022
(national CPI 10.0%)

+50bp

ECB's first hike
21 July 2022

record

energy & commodity prices
after the invasion

DNB, Financial Stability Report, spring 2022

"After a strong recovery from the pandemic, the economic outlook has worsened due to the war in Ukraine and high inflation, and financial stability risks have increased."

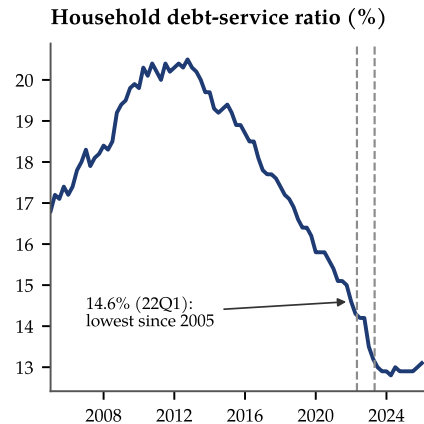
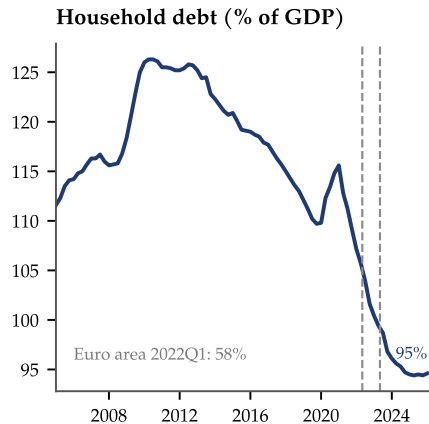
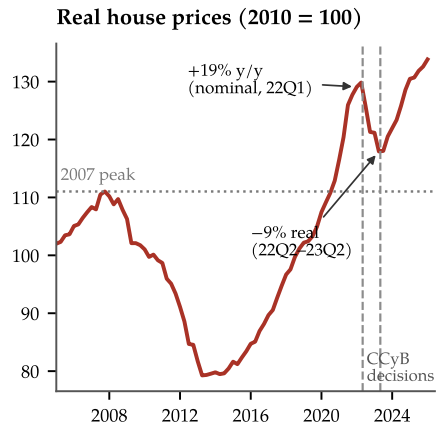
"Higher funding costs may put pressure on the debt sustainability of governments, businesses and households."

By end-2021 the Dutch economy was almost 3% larger than at end-2019. DNB's framework requires this recovery before the buffer is built.

Source: CBS (GDP, Nov 2021; inflation 2022); ECB monetary policy decision 21 Jul 2022; DNB Financial Stability Report, spring 2022.

In the financial system credit was cold and only housing was hot, and even housing soon turned

Q1 · ENVIRONMENT



Housing: nominal prices +19% y/y (22Q1), real 16% above the 2007 peak. DNB itself: *“the overheated housing market calls for further measures.”*

Credit: household debt high (107% of GDP vs 58% in the euro area) but *falling*; debt service at its lowest since 2005.

Source: BIS (WS_SPP, WS_TC, WS_DSR); DNB Financial Stability Report, spring 2022; own calculations. Gold lines: CCyB announcements (May-22, May-23).

Banks entered this storm well capitalised, and rising rates were lifting their profits

Q1 · ENVIRONMENT

17.7%

core capital (CET1) ratio
end-2021 (EU 15.7%, 21Q3)

16.3%

CET1 ratio, end-2022
“in line with EU average”

+6.7%

net interest income 2022
(67.7% of income)

11.5%

4 major banks' CET1 after a
severe stress test (min. 8%)

Q1 in one line: against raising now

war and energy shock · 11.6% inflation
rapid ECB tightening · house prices turning

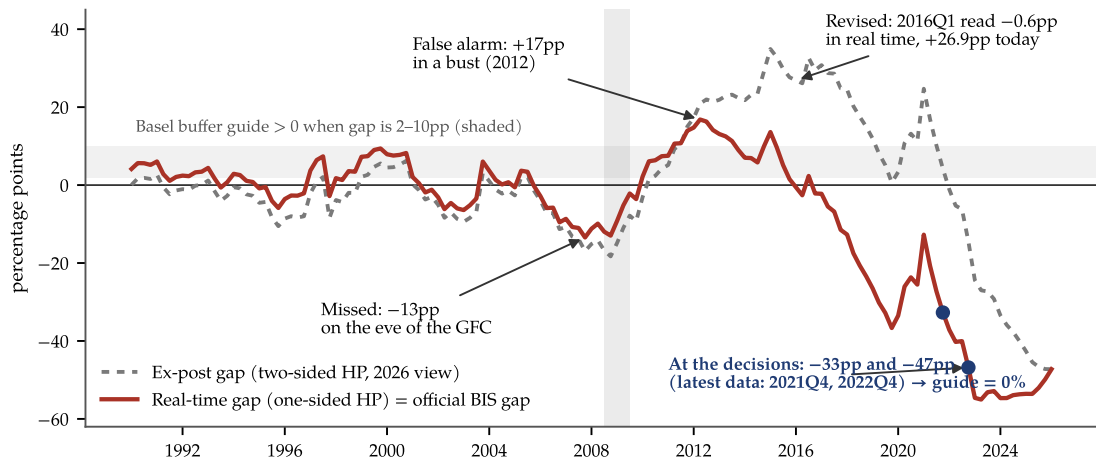
DNB (FSR 2022): higher rates “*may ultimately have a positive impact on the profitability of financial institutions.*”

... and for raising now

economy recovered · credit cold
banks strong, profits rising with rates

So why ignore the rulebook? On Dutch data the Basel rule had missed the last crisis and raised false alarms

Q2 · CRITERIA & RISKS

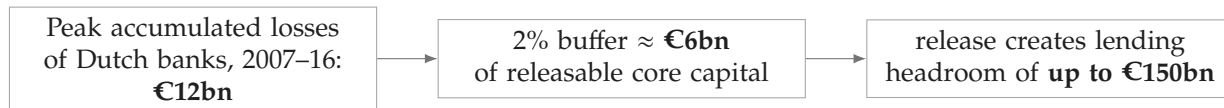


Credit-to-GDP gap = credit/GDP ratio minus its long-run trend (Hodrick-Prescott filter). **Rule:** gap < 2pp → 0%, gap > 10pp → 2.5%. Our one-sided filter ($\lambda = 400,000$) reproduces the official BIS gap within 0.00005pp (1971–2026).

Alternative trend filters are noisy around COVID (two-sided HP +4.2pp in 2021Q4), which is one more reason not to rely on any of them.

Source: BIS credit-to-GDP gaps (Q.NL.PA); Drehmann et al. (2010); Repullo & Saurina (2011); Edge & Meisenzahl (2011); own calculations.

DNB used its own compass instead: 2% in normal times, calibrated on past crisis losses rather than on credit growth



- Criteria: a dashboard (macro, banks, households & firms, markets); *“no mechanical link . . . guided discretion.”*
- 2% chosen *“also taking into account the impact of the buffer reduction in March 2020.”*
- Not alone: 17 jurisdictions run a *positive neutral* CCyB (kept filled in normal times); NL, Sweden, UK, Poland target 2%. ECB loss-based range: 1.1–1.8%.

Source: DNB (2022), Analytical framework for setting the CCyB in the Netherlands; BCBS (2024) d585, Table 1; De Nora, Pereira, Pirovano & Stammwitz (2025), ECB WP 3061.

Its stated reasons were recovery, uncertainty and the 2020 promise, never credit growth

Q2 · CRITERIA & RISKS

May 2022 · 0% → 1%

*“When the systemic risk buffers . . . were lowered in March 2020, we also announced our intention to **restore the buffers** by raising the CCyB.”*

“The risk profile is currently dominated by uncertainty caused by the war in Ukraine, but . . . the robust economic recovery . . . provides grounds for a gradual build-up.”

Cyclical risks: “normal” to elevated.

Both decisions: **a normal-to-elevated risk picture + strong banks + the promise**. Credit growth is never the reason.

May 2023 · 1% → 2%

*“Although the credit-to-GDP gap shows **no signs of excessive credit growth**, some other indicators already point to higher cyclical risk.”*

Cited: investors’ rising risk appetite · real-estate prices *falling* · debt sustainability of firms and governments · banks “robust, partly due to rising interest rates”.

The buffer was meant for shocks nobody can forecast, while the known hot spots got their own tools

Financial-stability risks named by DNB (FSR 2022/23)	Instrument
War, energy and supply shocks · rapid tightening of financial conditions and market corrections · debt sustainability of governments, firms, households · bank turmoil abroad (SVB, Credit Suisse, 2023)	CCyB, releasable for <i>any</i> shock
“Overheated” housing · high mortgage debt of first-time buyers · interest-only loans	borrower limits (loan-to-value, loan-to-income), tax reform, mortgage risk-weight floor (Jan 2022 – Nov 2026)
Size and concentration of the large banks (sector 400% → 280% of GDP)	buffer for systemically important banks (O-SII , cut 2024)

Principle: one instrument per target (Tinbergen, 1952)

DNB on the 2020 recomposition: more releasable capital is “a *valuable addition given the sensitivity/volatility of the Dutch economy to external events.*” A releasable buffer is the only tool that works whatever the source of the shock.

Source: DNB Financial Stability Reports, spring 2022 and spring 2023; ESRB O-SII notification NL (31 May 2023); DNB (24 Apr 2026); Tinbergen (1952).

Tested against the evidence, every prediction fits normalisation and none fits overheating

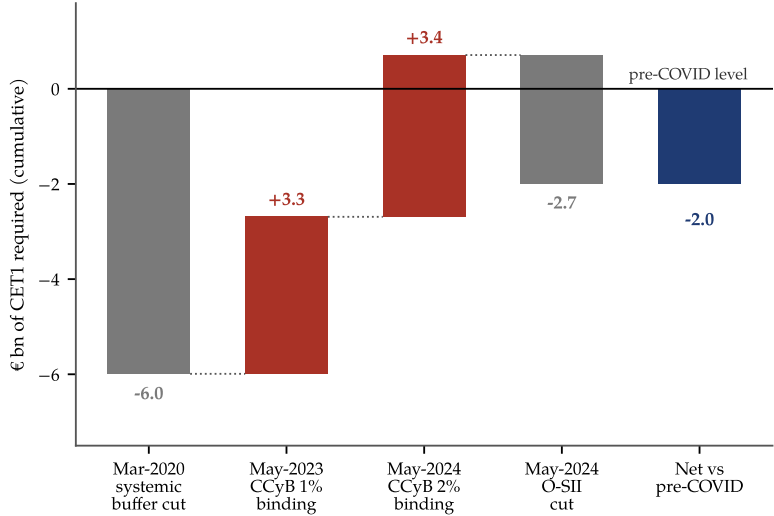
Q2 · CRITERIA & RISKS

If DNB were ...	normalising	fighting overheating	what happened	
Responding to credit indicators	no link	rises with them	gap, debt service, household debt ratio all <i>falling</i> ; CCyB <i>up</i>	✓
Path	+1pp a year, stop at 2%	follows risk	1% → 2% → held	✓
2023: house prices falling, rates jumping	raise anyway	pause	raised	✓
2024–26: house prices rising again (+10% y/y, Mar-26)	stay at 2%	go above 2% (framework phase 3)	held at 2%	✓
Other buffers	cut structural ones to compensate	keep them	cut in 2020 and 2024	✓

Five predictions, five matches: **normalisation**. The last row is the key to the whole story.

Source: BIS (WS_CREDIT_GAP, WS_DSR, WS_TC, WS_SPP); DNB press releases 2022–2026 (house prices: DNB, 26 Mar 2026); DNB framework (2022); ESRB O-SII notifications.

And the refill was a swap, not a squeeze: capital moved from structural buffers into the releasable one



DNB’s own words

“more or less capital-neutral for the three large banks” (2020)

“a limited increase in the net capital requirements” (Knot, Jun 2023)

“slightly increases the capital requirements for the banking sector as a whole” (FSR 2023)

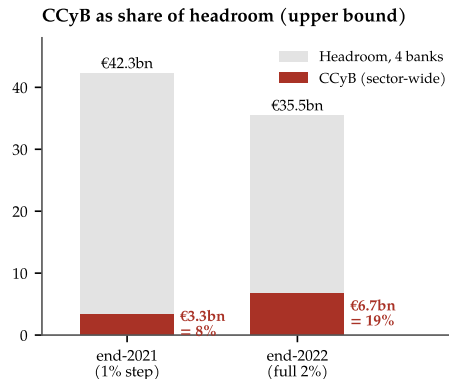
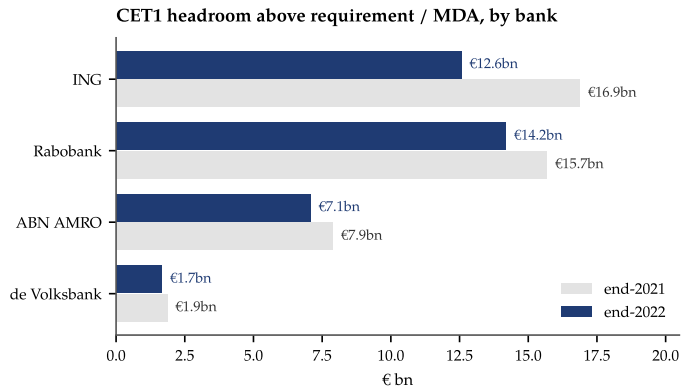
Not unusual: the UK offset its 2% neutral CCyB by lowering Pillar 2A and resolution requirements.

Waterfall: approximate, own calculation. Buffer cuts valued at end-2022 total RWA of ING, Rabobank, ABN AMRO, de Volksbank (BNG excluded);

CCyB amounts are DNB’s sector-wide figures on Dutch exposures. Relative to 2021 the change was an increase.
Source: DNB press release 17 Mar 2020; Knot (2023), statement to the Standing Committee for Finance, 7 Jun 2023; DNB FSR spring 2023; BCBS (2024) d585; bank annual reports 2022.

So was it the wrong moment? No. Higher rates gave banks the profits to build the buffer cheaply

Q3 · IMPLICATIONS



Headroom = CET1 capital above the bank's requirement (MDA trigger). Net interest income **+6.7%** in 2022 → buffers built from *retained earnings*, not by cutting loans. Headroom shrank because of RWA growth and payouts, not the CCyB.

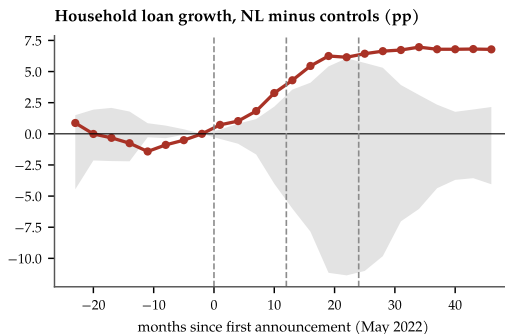
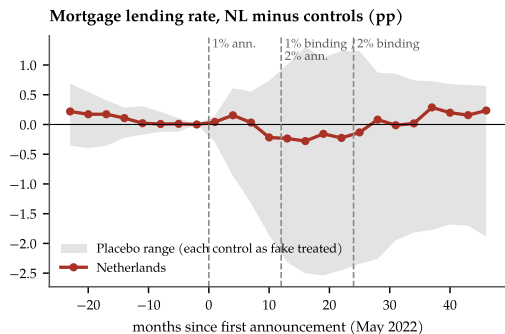
ECB: gradual build-up + high profitability “limit these economic costs” (MPB 24); activating buffers during the 2022–23 tightening “can mitigate implementation costs” (MPB 31).

It was not the wrong moment. Arguably it was the right one.

Source: Bank disclosures (ING, Rabobank, ABN AMRO, de Volksbank, 2021–22), de Volksbank requirement derived; DNB FSR 2023; ECB Macroprudential Bulletin 24 (2024) and 31 (2025, Box 2).

Borrowers did not pay either: credit kept flowing and its price moved with the rest of the euro area

Q3 · IMPLICATIONS



How to read: red = Netherlands minus countries that did not change their CCyB (AT, FI, IT, MT, LU); 0 = no difference; grey = differences that arise by chance (each control treated as if it had raised its CCyB).

- Dutch lending rates stayed **inside the placebo range** at all three CCyB dates
- Household credit grew **faster** than in peers (partly catch-up: NL lagged in 2019–22)

Source: ECB Data Portal (MIR, BSI); ESRB CCyB table (control selection); BIS (WS_TC); DNB press release 17 Sep 2026; own two-way fixed-effects event study.

DNB (Sep 2026): “no signs of constraints in bank credit supply.”

One treated country, small control group, housing rebound: no sign of contraction, not a causal estimate.

Nor did the buffer double the ECB's squeeze. It let monetary policy focus on inflation

Q3 · IMPLICATIONS

The concern

Two tightenings at once: ECB rates *and* capital requirements → credit crunch?

What the data show

Dutch lending rates did not diverge from euro-area peers (event study); bank profits *rose* with rates (net interest income +6.7%).

One monetary policy for the euro area, but national macroprudential policy: the CCyB is the **Dutch-specific lever** and the insurance against a rate shock gone wrong.

ECB Macroprudential Bulletin 31 (2025)

Early CCyB activation “helps monetary policy focus on its primary objective of price stability, thereby largely eliminating the potential for conflict between price stability and financial stability.”

DNB, FSR 2022

“A sudden tightening of financial conditions could also have a negative impact on financial stability.”

Source: ECB Macroprudential Bulletin 31 (Detken, Hempell & Pirovano, 2025); DNB Financial Stability Report, spring 2022.

Through reciprocity the 2% also binds foreign lenders, and the rest of Europe is moving the same way

Across borders

- Foreign banks with Dutch exposures must also hold the 2% (mandatory reciprocity up to 2.5%)
- Limits leakage to foreign lenders, a known problem with national capital tools (Aiyar, Calomiris & Wieladek, 2014); *non-bank* leakage remains
- In April 2023, 5 banking-union countries had positive-neutral frameworks; NL had the **highest** rate (2%)

CCyB increases in the euro area		
Country	Rate	Applies from
Germany	0.75%	Feb 2023
France	1.0%	Jan 2024
Ireland	1.5%	Jun 2024
Netherlands	2.0%	May 2024
Belgium	1.0%	Oct 2024
Portugal	0.75%	Jan 2026
Spain	1.0%	Oct 2026

Source: DNB press release 31 May 2023; ESRB CCyB rates table (Sep 2026); ECB Macropprudential Bulletin 21 (Behn et al., 2023); Aiyar, Calomiris & Wieladek (2014), JMCB.

€0 → €6.7bn

releasable-by-design capital

2020 → 2024

≈ half

of the €12bn peak losses

after the GFC

€150bn

lending headroom

if released

Asymmetric payoff (Lang & Menno, 2023)

Normal times: +1pp requirement → about **−0.1%** lending.

Bad times: 1pp release → up to about **+10%** lending.

Cheap to hold, powerful to release.

A growing role

The mortgage risk-weight floor expires on 30 Nov 2026.

DNB: this “*underpins the importance of . . . the current countercyclical capital buffer (CCyB) of 2%.*”

Credibility	Without a mechanical rule, markets must trust DNB's judgement and communication. This is the classic rules-vs-discretion trade-off (Kydland & Prescott, 1977).
Calibration	Why 2%, not 1.5%? The ECB's loss-based range is 1.1–1.8%; NL chose the top of the banking union.
Untested	No release yet. Will banks lend the freed capital, or hoard it as they did with usable buffers in COVID?
Baseline	Relative to 2021 (not 2019) it <i>was</i> an increase: +€3.3bn before the O-SII offset.
Same profits, two claims	The ECB lists the Netherlands among countries adopting bank levies on excess profits in 2022–23. These partly claim the earnings that fund the buffer.

Source: Kydland & Prescott (1977); De Nora et al. (2025); ECB Macroprudential Bulletin 21 (2023) and 31 (2025); DNB (2022, 2023).

In short, the promise was kept: the buffer was refilled in a storm, at the moment it was cheapest

EPILOGUE

Q1 · Environment

Recovered economy hit by war, inflation and rate hikes; credit cold; housing hot, then cooling; strong, profitable banks.

Q2 · Criteria & risks

2% normal-times level, calibrated on past losses; insurance against unforeseeable shocks; housing to other tools; structural buffers cut in return.

Q3 · Implications

Cheap · largely a swap · no sign of contraction · complements monetary policy · reciprocated abroad · €6.7bn releasable · relies on discretion.

In March 2020, the Netherlands had nothing to release.
Next time, it will.

The best time to fill a buffer is when nobody thinks you need it, even in a storm.

CCyB	Countercyclical capital buffer: 0–2.5% of risk-weighted assets on domestic exposures; raised with a 12-month lag, released immediately.	Credit-to-GDP gap	Credit/GDP ratio minus its long-run trend; the Basel reference for the CCyB (gap 2–10pp → buffer 0–2.5%).
Positive neutral CCyB	A CCyB kept above zero when risks are neither high nor low, so it can be released for any shock.	HP filter	Hodrick-Prescott filter: estimates a smooth trend; one-sided uses only past data (real time), two-sided also future data.
CET1	Common Equity Tier 1: the highest-quality capital (shares, retained earnings), as % of RWA.	Headroom	CET1 capital above a bank's requirement / MDA trigger.
RWA	Risk-weighted assets: assets weighted by their riskiness (e.g. mortgages count less than corporate loans).	Reciprocity	Foreign banks must apply the Dutch CCyB rate to their Dutch exposures.
Pillar 1 / P2R / P2G	Legal minimum (4.5% CET1) / bank-specific add-on set by the supervisor / non-binding supervisory guidance.	Borrower-based measures	Limits on loan-to-value (LTV) and loan-to-income (LTI) of new mortgages.
CCoB	Capital conservation buffer, 2.5% for all banks.	Risk-weight floor	Minimum average risk weight on Dutch mortgages of banks using internal models (Art. 458 CRR), Jan 2022 – Nov 2026.
O-SII / SRB	Buffer for other systemically important institutions / systemic risk buffer (structural, not designed for release).	Event study / placebo	Compare NL with countries that did not change their CCyB around each date / repeat with each control as a fake “treated” country.
MDA	Maximum distributable amount: if CET1 falls below the combined buffer requirement, payouts are automatically restricted.	pp / bp	Percentage point / basis point (0.01pp).

Methods. Credit gap: one-sided (recursive) and two-sided HP filter, $\lambda = 400,000$, sample from 1961Q1; Hamilton (2018) regression filter, $h = 8, 20, p = 4$. Headroom: (CET1 ratio — CET1 requirement/MDA) $\times$ total RWA per bank; CCyB amount sector-wide (upper bound); 6–47% assumption grid as sensitivity. Event study: two-way FE, NL $\times$ 3-month event-time bins, country and month FE; placebo inference over controls with no CCyB change 2021–25. Code and data: GitHub repository (code/, data/).

Primary sources

BCBS (2024). Range of practices in implementing a positive neutral CCyB (d585).
 DNB (2020). DNB lowers bank buffer requirements to support lending, 17 Mar.
 DNB (2022). Analytical framework for setting the CCyB in the Netherlands.
 DNB (2022, 2023). CCyB press releases, 27 May 2022, 31 May 2023; O-SII news item, 31 May 2023.
 DNB (2026). CCyB decisions 26 Mar and 17 Sep; risk-weight measure expires, 24 Apr.
 DNB. Financial Stability Report, spring 2022 and spring 2023.
 Knot, K. (2023). Introductory statement, Standing Parliamentary Committee for Finance, 7 Jun.
 ECB (2020). Temporary capital and operational relief, 12 Mar; ECB (2022). Monetary policy decisions, 21 Jul.
 ESRB. CCyB rates table; O-SII notifications NL (27 Nov 2020; 31 May 2023); COVID-19 measures.
 Data: BIS (credit gaps, total credit, DSR, property prices); ECB Data Portal (MIR, BSI); CBS; bank annual and Pillar 3 reports (ING, Rabobank, ABN AMRO, de Volksbank).

Literature

Aiyar, Calomiris & Wieladek (2014). Does macro-prudential regulation leak? *JMCB* 46(s1).
 Behn, Pereira, Pirovano & Testa (2023). A positive neutral rate for the CCyB. *ECB MPB* 21.
 Couaillier, Lo Duca, Reghezza & Rodriguez d’Acri (2022). Caution: do not cross! *ECB WP* 2644.
 De Nora, Pereira, Pirovano & Stammwitz (2025). From losses to buffer. *ECB WP* 3061.
 Detken, Hempell & Pirovano (2025). Macroprudential and monetary policy interaction. *ECB MPB* 31.
 Drehmann, Borio, Gambacorta, Jiménez & Trucharte (2010). Countercyclical capital buffers: exploring options. *BIS WP* 317.
 Edge & Meisenzahl (2011). The unreliability of credit-to-GDP ratio gaps in real time. *IJC* 7(4).
 Hamilton (2018). Why you should never use the Hodrick-Prescott filter. *REStat* 100(5).
 Herrera, Scalone & Pirovano (2024). The importance of being positive. *ECB MPB* 24.
 Kydland & Prescott (1977). Rules rather than discretion. *JPE* 85(3).
 Lang & Menno (2023). The state-dependent impact of changes in bank capital requirements. *ECB WP* 2828.
 Repullo & Saurina (2011). The countercyclical capital buffer of Basel III: a critical assessment. *CEMFI WP* 1102.
 Tinbergen (1952). *On the Theory of Economic Policy*. North-Holland.